

February 1996

xxxinv - Transforms input Easting and Northing to longitude and latitude for the xxx projection

SYNTAX

FUNCTION xxxinv (x, y, lon, lat)

```
double x;  
double y;  
double *lon;  
double *lat;
```

PARAMETERS

x (input, double)

Easting value. Value given in meters.

y (input, double)

Northing value. Value given in meters.

lon (output, double)

Output longitude. Value given in radians.

lat (output, double)

Output latitude. Value given in radians.

DESCRIPTION

xxxinv() converts input Easting and Northing to the corresponding longitude and latitude for the xxx projection (xxx is replaced by the abbreviation of the projection). The following subroutines follow this general format:

utminv(x, y, lon, lat)	-- Universal Transverse Mercator (UTM)
stplninv(x, y, lon, lat)	-- State Plane
alberinv(x, y, lon, lat)	-- Albers
lamccinv(x, y, lon, lat)	-- Lambert Conformal Conic
merinv(x, y, lon, lat)	-- Mercator
psinv(x, y, lon, lat)	-- Polar Stereographic

<code>polyinv(x, y, lon, lat)</code>	-- Polyconic
<code>eqconinv(x, y, lon, lat)</code>	-- Equidistant Conic
<code>tminv(x, y, lon, lat)</code>	-- Transverse Mercator (TM)
<code>sterinv(x, y, lon, lat)</code>	-- Stereographic
<code>lamazinv(x, y, lon, lat)</code>	-- Lambert Azimuthal
<code>aziminv(x, y, lon, lat)</code>	-- Azimuthal Equidistant
<code>gnominv(x, y, lon, lat)</code>	-- Gnomonic
<code>orthinv(x, y, lon, lat)</code>	-- Orthographic
<code>gvnspinv(x, y, lon, lat)</code>	-- General Near Side Perspective
<code>sininv(x, y, lon, lat)</code>	-- Sinusoidal
<code>equiinv(x, y, lon, lat)</code>	-- Equirectangular
<code>millinv(x, y, lon, lat)</code>	-- Miller
<code>vandginv(x, y, lon, lat)</code>	-- Van der Grinten
<code>omerinv(x, y, lon, lat)</code>	-- Hotine Oblique Mercator (HOM)
<code>robinv(x, y, lon, lat)</code>	-- Robinson
<code>sominv(x, y, lon, lat)</code>	-- Space Oblique Mercator (SOM)
<code>alconinv(x, y, lon, lat)</code>	-- Alaska Conformal
<code>goodinv(x, y, lon, lat)</code>	-- Goode
<code>molwinv(x, y, lon, lat)</code>	-- Mollweide
<code>imolwinv(x, y, lon, lat)</code>	-- Interrupted Mollweide
<code>haminv(x, y, lon, lat)</code>	-- Hammer
<code>wivinv(x, y, lon, lat)</code>	-- Wagner IV
<code>wviiinv(x, y, lon, lat)</code>	-- Wagner VII
<code>obleginv(x, y, lon, lat)</code>	-- Oblated Equal Area

init_inv() must be called once before calling any of the above subroutines. For example, if *utminv()* is to be called, *init_inv()* must be called setting the UTM projection, UTM zone, and ellipsoid. Then *utminv()* may be called as many times as needed even if *init_inv()* is called initializing another projection. The only exception is if *init_inv()* is called initializing UTM with another zone or ellipsoid, then *init_inv()* must be called again to perform the original conversion.

RETURN VALUE

<code>E_SUCC</code>	<code>(0)</code>	--> Successful completion
<code>ERROR</code>	<code>(>0)</code>	--> An error was encountered
<code>IN_BREAK</code>	<code>(-2)</code>	--> Coordinates fell into interrupted area

SPECIAL CONSIDERATIONS:

The *xxxinv()* call is for C only.